Academic Year: 2025-2026

Faculty: Caryn Geady, PhD

Program Coordinator: Andres Felipe Melani De La Hoz, MSc

Associate Dean: Erin Mandel-Shorser, PhD

Schedule Type: LEC

Course Outline

Course Name: ***BINF 5507-Machine Learning and AI for Bioinformatics***

Land Acknowledgement

Humber College is located within the traditional and treaty lands of the Mississaugas of the Credit. Known as Adoobiigok [A-doe-bee-goke], the "Place of the Black Alders" in Michi Saagiig [Mi-Chee Saw-Geeg] language, the region is uniquely situated along Humber River Watershed, which historically provided an integral connection for Anishinaabe [Ah-nish-nah-bay], Haudenosaunee [Hoeden-no-shownee], and Wendat [Wine-Dot] peoples between the Ontario Lakeshore and the Lake Simcoe/Georgian Bay regions. Now home to people of numerous nations, Adoobiigok continues to provide a vital source of interconnection for all.
For more information, visit the Aboriginal Resource Centre (LRC2137) North Campus, (WEL301) Lakeshore Campus or www.humber.ca/aboriginal/

Faculty	Faculty of Health Sciences and Wellness
Program	Graduate Certificate in Clinical Bioinformatics
Course Name:	BINF 5507: Machine Learning and AI for Bioinformatics
Pre-Requisite(s)	BINF 5007
Co-Requisite(s)	None

Pre-Requisite(s) for	None
Equates	None
Restrictions	Clinical Bioinformatics Students
Credit Value	4
Total Course Hours	56

Developed by:

Approved by:

Course Description

This course provides learners with an introduction to the principles and applications of machine learning and artificial intelligence in the field of clinical bioinformatics. This course provides the theoretical and hand-on training of the fundamental concepts of machine learning along with machine learning algorithms, and their applications in medical imaging and genomics.

Course Rationale

This course provides learners with essential skills in supervised and unsupervised techniques like classification, clustering, and dimensionality reduction. Through hands-on labs, students will learn to preprocess data, build predictive models, and uncover patterns in data. Learners will gain experience with specialized techniques for survival analysis and time-to-event modeling, crucial for clinical trials and patient prognosis. By the end of the course, students will be equipped to apply machine learning models to real-world bioinformatics problems, ensuring they are prepared for roles in healthcare data analysis, pharmaceutical R&D, and clinical decision support systems.

Program Learning Outcomes Emphasized in this Course:

1. Identify relevant evidence-based research, trends, technologies, and standards within the field of bioinformatics to maintain currency in the industry.
2. Apply bioinformatics and computational methods to the design, execution and interpretation of scientific research.
3. Employ programming language and scripting versatility to be able to automate processes at the application level.
4. Examine bioinformatics data to discover patterns, draw conclusions and generate predictions for subsequent experiments.
5. Comply with legal obligations, as well as with the professional and ethical standards that ensure privacy, security and confidentiality in the access, retention, storage and disposal of personal genetic data and related health information.
6. Use statistics in the analysis, collation and interpretation of data to make informed decisions.

Course Format(s)

Lecture
Tutorial

Course Learning Outcomes

OQF Category	At the successful completion of the course, the student will have demonstrated an ability to:
Depth and Breadth of Knowledge	1. Critically assess the fundamental concepts of machine learning and their relevance to clinical bioinformatics. 2. Analyze various machine learning models to determine their applicability to different types of clinical bioinformatics problems.
Knowledge of Methodologies	3. Describe methodologies for preprocessing and handling clinical bioinformatics data to allow for improved biomarker discovery. 4. Distinguish between different types of machine learning algorithms to justify the selection of appropriate machine learning techniques for specific clinical bioinformatics problems.
Application of Knowledge	5. Develop end-to-end pipelines including data preprocessing, model training, and evaluation to address clinical bioinformatics problems
Communication Skills	6. Demonstrate the ability to articulate concepts to both technical and non-technical audiences through reports, presentation and defense of (group) findings on bioinformatics challenges.
Professional Capacity/ Autonomy	7. Demonstrate the ability to independently design and implement machine learning projects in clinical bioinformatics for relevant applications.

Assessment Weighting

Assessment	Weight
GitHub Integration / Participation	10%
Assignments	40%
Midterm Exam	15%
Capstone Project	35%
Total	**100%**

Modules of Study

Module	Course Learning Outcomes	Assessments
Introduction to Machine Learning and AI for Biomarker Discovery and the Joys of Data Wrangling	1-3, 5	GitHub Integration Coding Assignment Midterm Test
Algorithms I: Supervised Learning	1-6	GitHub Integration Coding Assignment Midterm Test
Algorithms II: Unsupervised Learning and Survival Analysis	1-6	GitHub Integration Coding Assignment Midterm Test
Applications of ML in Pharmacogenomics and Radiomics	3-7	GitHub Integration
Capstone Project / Independent Study	3-7	Mini-report Oral presentation

Required Resources, Tools and/or Equipment:

- None.

Supplemental Resources:

- Blackboard lectures and assigned resources

Additional Tools and Equipment

- Blackboard lectures and assigned resources

Prior Learning Assessment and Recognition (PLAR)

Students who have prior learning in the material of this course may be eligible for a course credit in recognition of their prior learning. The following table indicates the method that is used to assess prior learning for this course, or it indicates that such an assessment is not available. Students must apply for consideration for a prior learning assessment through the Office of the Registrar, and there is usually a fee associated with the application.

Portfolio	Challenge Exam	Skills Test	Interview	Other (Specify)	Not Available For PLAR
X	☐	☐	☐	☐	☐

Policies and Procedures

It is the student's responsibility to be aware of their obligations under Humber Policies and Procedures.

Academic Regulations

It is the student's responsibility to be aware of the College Academic Regulations. The Academic Regulations apply to all applicants to Humber and all current students enrolled in any program or course offered by Humber, in any location. Information about **academic appeals** is found in the Academic Regulations.

Accessible Learning Services

Humber strives to create a welcoming environment for all students where equity, diversity and inclusion are paramount. Accessible Learning Services facilitates equal access for students with disabilities by coordinating academic accommodations and services. Staff in Accessible Learning Services are available by appointment to assess specific needs, provide referrals and arrange appropriate accommodations. If you require academic accommodations, contact:

Accessible Learning Services: http://www.humber.ca/student-life/swac/accessible-learning
North Campus: (416) 675-6622 X5090
Lakeshore Campus: (416) 675-6622 X3331

Academic Integrity

Academic integrity is essentially honesty in all academic endeavors. Academic integrity requires that students avoid all forms of academic misconduct or dishonesty, including plagiarism, cheating on tests or exams or any misrepresentation of academic accomplishment.

Disclaimer

While every effort is made by the professor/faculty to cover all material listed in the outline, the order, content, and/or evaluation may change in the event of special circumstances (e.g. time constraints due to inclement weather, sickness, college closure, technology/equipment problems or changes, etc.). In any such case, students will be given appropriate notification.

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